

Enterprise Demand Forecasting & Inventory Decision Support System

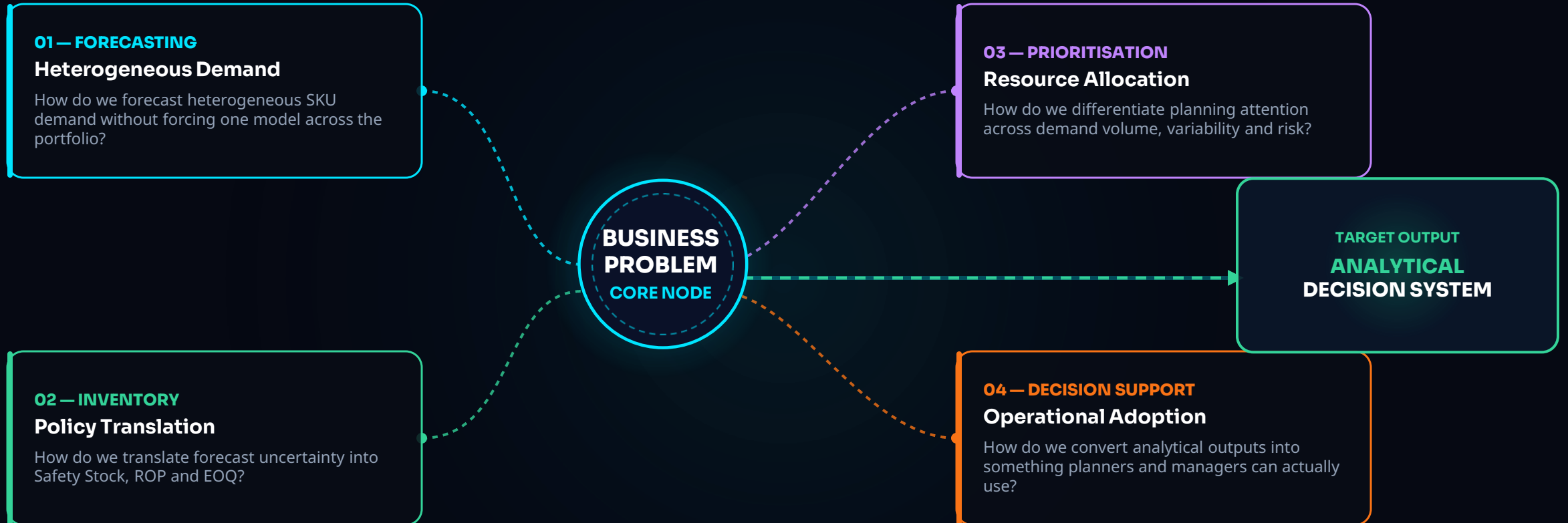
Using Time Series Analysis

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The Business Problem We Set Out to Solve

Four business questions define the analytical challenge.



Objective: Build a repeatable SKU-level decision-support system connecting historical demand to forecast, inventory and management decisions.

From Historical Demand to Supply Chain Decisions



"The objective is not simply to forecast demand — it is to convert demand history into better supply-chain decisions."

PORTFOLIO SCALE

1,401 SKUs

SOLUTION ARCHITECTURE

Forecasting

SKU-level demand intelligence

33,648 Transaction Rows

Inventory

Safety Stock + ROP + EOQ

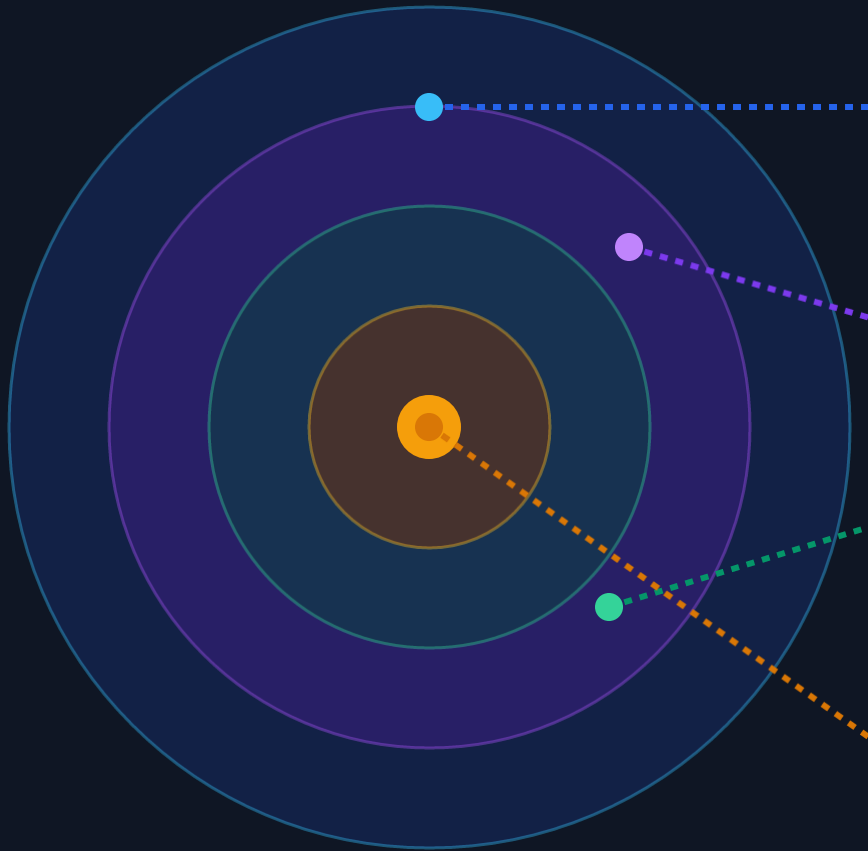
24 Months of History

Decision Intelligence

Risk + ABC-XYZ + Financial visibility

The Challenge Is Not Just Forecasting

Four interconnected business dimensions that must be solved together:



DEMAND

Different SKUs, different patterns

FORECASTING

Uncertainty varies by SKU

INVENTORY

Balance service vs. capital

BUSINESS

Every planning decision has financial consequences

A forecast creates value only when it improves the next decision.

Moving Beyond One-Size-Fits-All Planning

Seven distinct demand structures — each requiring SKU-level forecasting evaluation

TRADITIONAL PLANNING

- 01 Generic Forecast**
Top-down, aggregated demand estimates
- 02 Generic Buffer**
Uniform safety stock percentage across all SKUs
- 03 Generic Replenishment**
Standard reorder rules without variance intelligence

PROJECT APPROACH

- 01 SKU-Level Diagnostics**
Granular item-level demand profiling and behavior classification
- 02 Model Competition**
Selection of the best validated forecasting model per SKU
- 03 Forecast + Safety Stock**
Dynamic buffer calculation paired with tailored demand signals
- 04 ABC-XYZ + Risk Intelligence**
Multi-dimensional prioritization using demand and risk signals

FROM GENERIC PLANNING → TO EVIDENCE-DRIVEN DECISIONS

Demand Does Not Behave the Same Way Across SKUs

Seven distinct demand structures — classified via explicit rule-based decision logic:

DEAD STOCK

Zero-demand history

Rule: $\text{Mean} = 0$



INTERMITTENT

Sporadic demand with zeros

Rule: $\text{Zero\%} \geq 50\%$ & $\text{ADI} \geq 2$



LUMPY

Irregular high-magnitude spikes

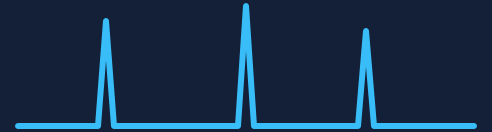
Rule: $\text{ADI} \geq 1.32$ & $\text{CV}^2 \geq 0.49$



SEASONAL

Repeating calendar patterns

Rule: $\text{Lag12} \geq 0.50$



TRENDING

Directional growth or decline

Rule: $|\text{Slope}| \geq 0.05 \times \text{Mean}$



HIGHLY VOLATILE

Extreme variation, no pattern

Rule: $\text{CV} \geq 1$



STABLE

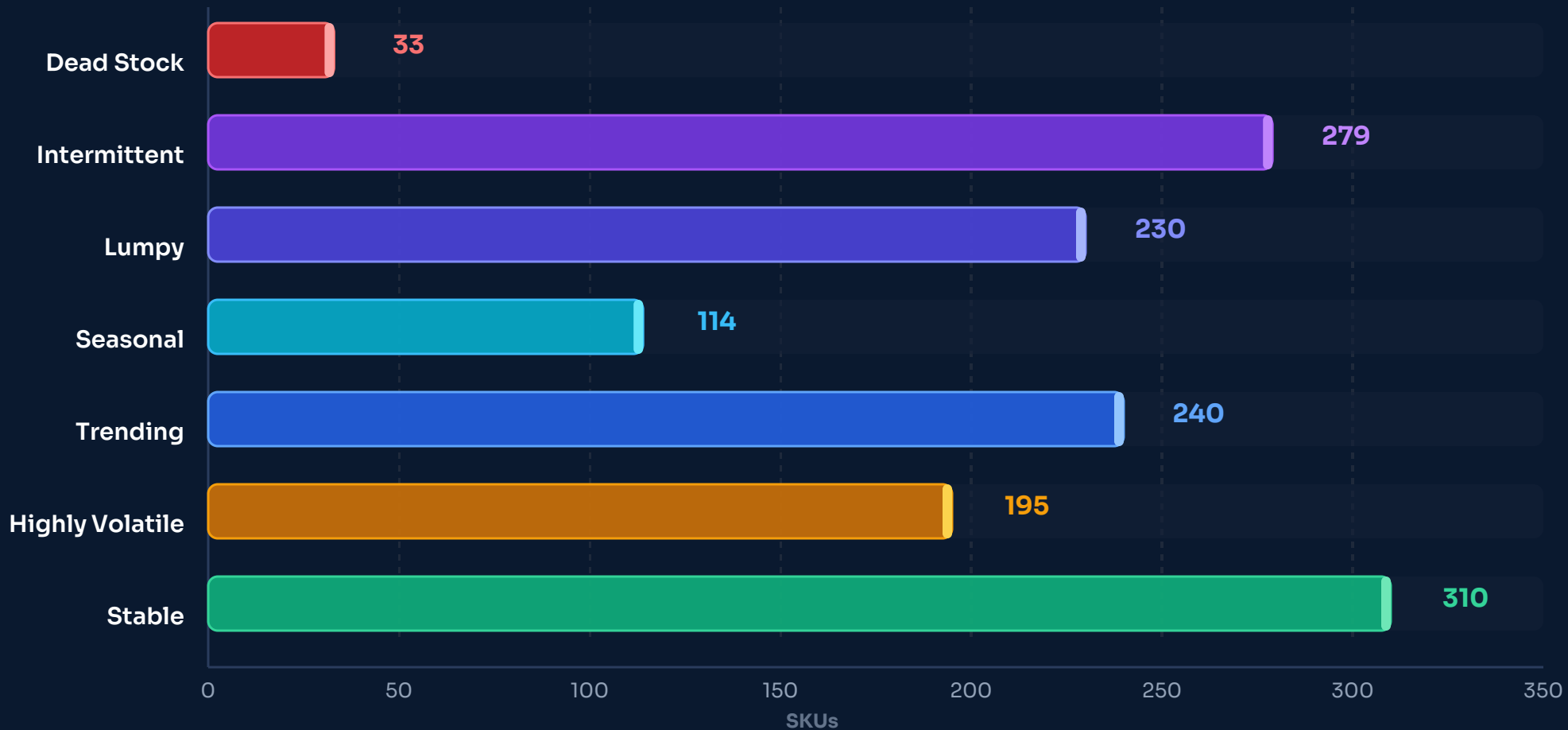
Consistent, predictable demand

Rule: $\text{CV} < 1$



The Portfolio Is Structurally Heterogeneous

Demand-pattern distribution across **1,401 SKUs** — demonstrating why no single forecasting approach fits the portfolio.



Demand heterogeneity is the reason the project evaluates forecasting at SKU level.

Rule-based classification uses Mean, CV, Zero %, ADI, CV², trend slope and lag-12 correlation with explicit thresholds.

24 Months of Enterprise Demand → Decision-Ready Analytical Dataset

End-to-end data pipeline transforming raw monthly files into an enterprise analytical dataset:



Data integrity is part of the model — not an afterthought.

Raw Data Snapshot — DATA-ML.xlsx

SKU-level inventory movement data across 24 monthly worksheets

SOURCE: DATA-ML.xlsxWORKSHEET: APRIL24

Particulars	Opening	Inwards	Outwards	Closing
101 Beige-L (T-Shirt)	—	330 pcs	3 pcs	327 pcs
101 Beige-M (T-Shirt)	—	330 pcs	3 pcs	327 pcs
101 Beige-S (T-Shirt)	—	165 pcs	—	165 pcs
101 Beige-XL (T-Shirt)	—	205 pcs	3 pcs	202 pcs
101 Beige-XXL (T-Shirt)	—	102 pcs	2 pcs	100 pcs
101 Beige-XXXL (T-Shirt)	—	12 pcs	—	12 pcs
101 B.Green T-Shirt:-L	—	1,090 pcs	42 pcs	1,048 pcs
... 33,648 total transaction rows across 1,401 SKUs ...				

24 MONTHLY SHEETS

APR 2024 → MAR 2026

Worksheets consolidated to capture demand variation.

SKU-LEVEL RAW INPUT

Particulars + Movement Fields

Transactional log of inventory movements.

PRE-ANALYTICS STATE

Raw Operational Records

Raw data prior to ML forecasting pipelines.

Raw operational data becomes the foundation for the analytical pipeline.

From Raw Demand to Inventory Decision

From data preparation to validated forecasting and inventory decision support.

PHASE 1: INGEST & DIAGNOSE

01cleaning_
services

CLEAN

Prepare historical transactions

Input: Raw Monthly Operational Spreadsheets

02analytics

DIAGNOSE

Inspect demand patterns and issues

ADI & CV² Metrics

PHASE 2: CANDIDATE EVALUATION & SELECTION

03account_
tree

MODEL

Build candidate forecasting models

20 Months Train

04fact_che
ck

VALIDATE

Evaluate held-out performance

4 Months Unseen

05emoji_ev
ents

SELECT WINNER

Choose the best model

Lowest sMAPE

PHASE 3: FINAL FORECAST & DECISION

06autorene
w

REFIT & FORECAST

Retrain on full data and forecast demand

Full 24M Dataset

07inventor
y_2

INVENTORY DECISION

Translate forecast into planning action

Actionable Plan

End-to-end governance: held-out validation performance drives model selection and inventory positioning.

From Forecast to Inventory Decision

The forecast becomes an inventory-control input.

INVENTORY INTELLIGENCE ENGINE

1. FORECAST

Next-period demand forecast

2. RMSE

Quantifies forecast error

3. LEAD-TIME SCALING

Scales RMSE across fixed 7-day lead time

4. SAFETY STOCK

Quantified Protection Buffer
Converts forecast uncertainty into an explicit stock buffer.

CONTROL PARAMETERS (ASSUMPTIONS)

7 DAYS

Fixed Lead Time
Replenishment horizon

1.645

Service-Level Z
Model assumption

FROM UNCERTAINTY TO CONTROL

Higher RMSE → Higher Safety Stock

FORECAST

..

FORECAST
ERROR

..

PROTECTION
BUFFER

..

PLANNING
DECISION

SERVICE

Protection

INVENTORY

Positioning

MANAGEMENT

Visibility

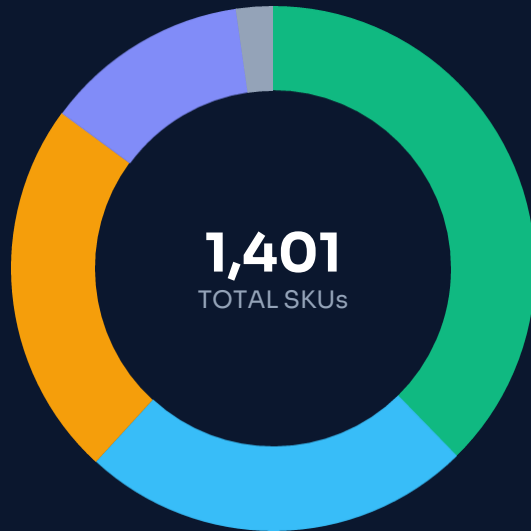
INTERPRETATION

Forecast uncertainty becomes a measurable inventory-control input. That uncertainty feeds Safety Stock, Reorder Point and EOQ decisions.

Enterprise Model Competition — 1,401 SKUs

SKU-level model competition shows why a single forecasting method cannot serve the entire portfolio.

MODEL COMPETITION WINNERS DISTRIBUTION



1,401 SKUs evaluated; shares shown as a percentage of the portfolio.

Enterprise Portfolio Accuracy

1,401 SKUs evaluated

Mean Validation Accuracy: **62.84%** | Median: **63.28%**

Accuracy is project-defined and derived from validation sMAPE.

FORECAST RISK DISTRIBUTION

LOW RISK

43

SKUs

MEDIUM RISK

246

SKUs

HIGH RISK

1,112

SKUs

KEY TAKEAWAY

The portfolio contains SKU-specific winners — no single forecasting model is applied to everything.

Safety Stock — Buffering Forecast Uncertainty

Convert forecast error into a quantified protection buffer.

MATHEMATICAL CONVERSION ENGINE

1. FORECAST ERROR (RMSE)

RMSE = forecast error metric | Input uncertainty signal

↓ Scale across lead time: $\times \sqrt{(7 / 30)}$

2. LEAD-TIME SCALING (σ_{LT})

$$\sigma_{LT} = \text{RMSE} \times \sqrt{(7/30)}$$

Scales RMSE across the fixed 7-day lead time

↓ Apply Service-Level Z multiplier: $\times 1.645$

3. SAFETY STOCK (SS)

$$\text{Safety Stock} = 1.645 \times \sigma_{LT}$$

Quantified protection buffer derived from forecast error

MATHEMATICAL RELATIONSHIP

HIGHER RMSE



HIGHER LEAD-TIME-SCALED UNCERTAINTY



HIGHER SAFETY STOCK

Forecast error directly expands the uncertainty buffer required for inventory protection.

SPECIAL CASE / ZERO VARIANCE DESIGN:

Dead-stock / constant-demand cases → Safety Stock = 0 by design.

Safety stock is the quantified buffer against forecast uncertainty.

Reorder Point & EOQ — When to Reorder, How Much to Order

Forecast + uncertainty determine the replenishment point; EOQ determines the order quantity.

DECISION ENGINE 1: WHEN TO REORDER (ROP)

INPUT DRIVERS: DEMAND & SAFETY BUFFER

$$\text{LTD} = (\text{Forecast} / 30) \times 7 \mid \text{ROP} = \text{LTD} + \text{Safety Stock}$$

Converts monthly forecast into 7-day lead-time demand and adds the safety buffer.

Combine expected demand + safety buffer

OUTPUT THRESHOLD

ROP

LTD + Safety Stock

Triggers order placement at ROP level

DECISION ENGINE 2: HOW MUCH TO ORDER (EOQ)

INPUT DRIVERS & WILSON EOQ FORMULA

$$\text{EOQ} = \sqrt{(2 \times D \times S / H)}$$

Balances Demand (D), Ordering Cost (S), and Holding Cost (H) under the model assumptions.

Output: EOQ order quantity under the model assumptions

OUTPUT POLICY

EOQ

Optimal Order Batch

Balances ordering and holding-cost trade-offs

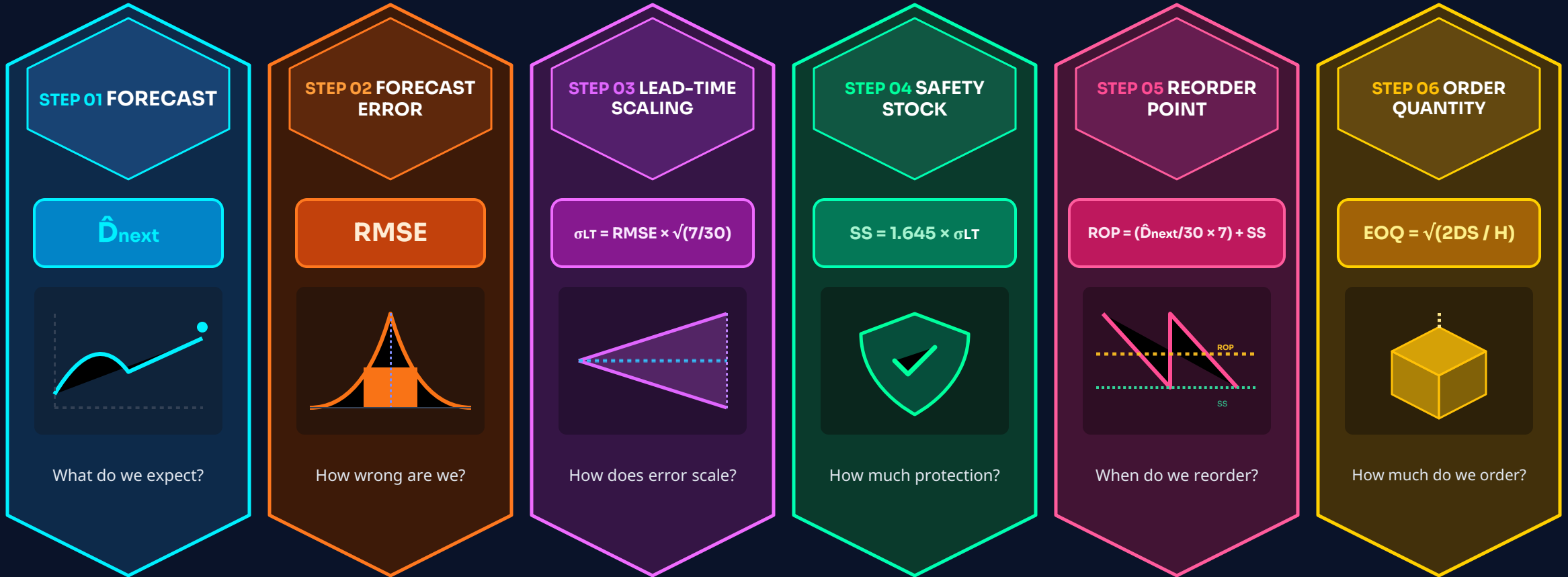
CONTROL PARAMETERS (MODEL ASSUMPTIONS):

Lead Time = 7 days | Ordering Cost = 500/event | Holding Cost = 3% of unit value | Unit Cost = 250/unit

ROP answers: **WHEN SHOULD WE REORDER?** | EOQ answers: **HOW MUCH SHOULD WE ORDER?**

Mathematical Execution — From Forecast Error to Inventory Policy

The model connects forecast error to inventory-policy calculations.



Forecast uncertainty is translated into measurable inventory policy.

Decision Support Dashboard

From model outputs to interactive management visibility

WHAT IT ENABLES

01 Portfolio Visibility

Demand, forecast & risk in one place.

02 Drill-Down Intelligence

Seamless flow to segment & SKU level.

03 Auditability & Control

Traceable analytical decision records.

Python Models → Audit → Dashboard

ENTERPRISE DECISION SUPPORT SYSTEM — EXECUTIVE DASHBOARD

01 — EXECUTIVE CONTROL TOWER

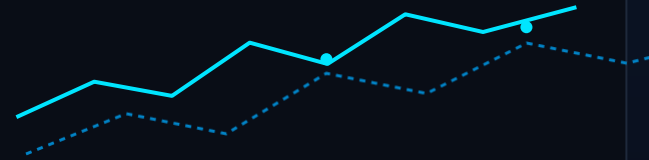
PORTFOLIO OVERVIEW
Active SKU Health

FORECAST ACCURACY
Model Performance

INVENTORY POSITION
Safety Stock Health

RISK EXPOSURE
Exception Diagnostics

02 — DEMAND & FORECAST INTELLIGENCE



Demand Patterns • Model Competition • Selection

03 — INVENTORY OPTIMISATION

SAFETY STOCK
Buffer Policy

REORDER POINT
ROP Trigger

EOQ
Batch Size

Model-Derived Policy Parameters

04 — RISK & SEGMENTATION



ABC-XYZ Matrix
Portfolio → SKU Drill Down

Forecast Risk & Demand Diagnostics Visibility

05 — FINANCIAL & AUDIT VISIBILITY

INVENTORY VALUE
Capital Allocation

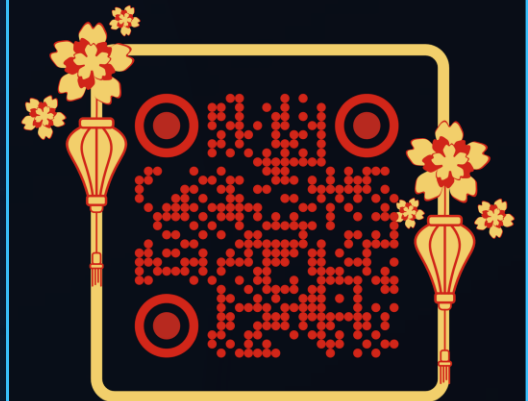
CARRYING COST
Holding Exposure

STOCKOUT RISK
Service Exposure

Model-Derived Financial Visibility & Full Traceability

LIVE DEMO

Interactive Dashboard



SCAN TO EXPLORE DASHBOARD

Dashboard = Interactive Decision-Support Layer

LIVE DASHBOARD DEMONSTRATION

Portfolio-Level Model Outputs

Portfolio-level results from the forecasting and inventory decision framework

1. FORECAST & SERVICE

AVERAGE FILL RATE

99.87%

Portfolio mean fill rate

AVERAGE SERVICE LEVEL

94.38%

Portfolio mean service level

2. INVENTORY POLICY

TOTAL SAFETY STOCK

37,049 units

TOTAL REORDER POINT (ROP)

48,408 units

TOTAL EOQ (ECONOMIC ORDER QTY)

275,481 units

Mean Inventory Turnover: **~2.40×**

Portfolio-Implied Inventory Days: **~151.89 days**

Labelled as **MODEL-IMPLIED / PROJECT ANALYTICS** values.

3. FINANCIAL VISIBILITY

MODEL-IMPLIED INVENTORY VALUE

43,697,375

Model-implied analytical measure

ESTIMATED CARRYING COST

1,310,921.25

ESTIMATED STOCKOUT COST / EXPOSURE

610,938.02

Model-implied analytical measures, not accounting outcomes. Realised operational and financial impact requires deployment and post-deployment measurement.

The model converts SKU-level forecast behaviour into **quantified portfolio-level inventory policy**.

ABC-XYZ Segmentation — Prioritise the Portfolio

Not every SKU deserves the same planning attention.

ABC: Demand Volume XYZ: Demand Variability	X Lower Demand Variability	Y Medium Demand Variability	Z Higher Demand Variability	Total
A Higher Volume Contribution	0	22	455	477
B Medium Volume Contribution	0	0	369	369
C Lower Volume Contribution	32	0	523	555
Total	32	22	1,347	1,401

RESULT: 1,347 of 1,401 SKUs are classified as Z under the project's CV-based variability classification.

IMPLICATION: The portfolio shows substantial demand variability, supporting differentiated rather than uniform planning logic.

From Analytical Results to Departmental Action

How portfolio findings translate into functional decision priorities

FORECASTING / PLANNING

Use SKU-level forecasts and demand diagnostics to set differentiated planning frequency.

PROCUREMENT

Use ROP, EOQ and risk signals to support replenishment and supplier-planning decisions.

WAREHOUSE

Use safety stock and risk segmentation to prioritise inventory attention and exceptions.

ANALYTICAL DECISION SUPPORT

INTELLIGENCE CORE

FINANCE

Use inventory value, carrying cost and stockout exposure for working-capital visibility.

MANAGEMENT

Use portfolio-level KPIs, risk and segmentation for prioritised decision-making.

One analytical system, multiple functional decisions.

Realised business impact requires operational deployment and measurement.

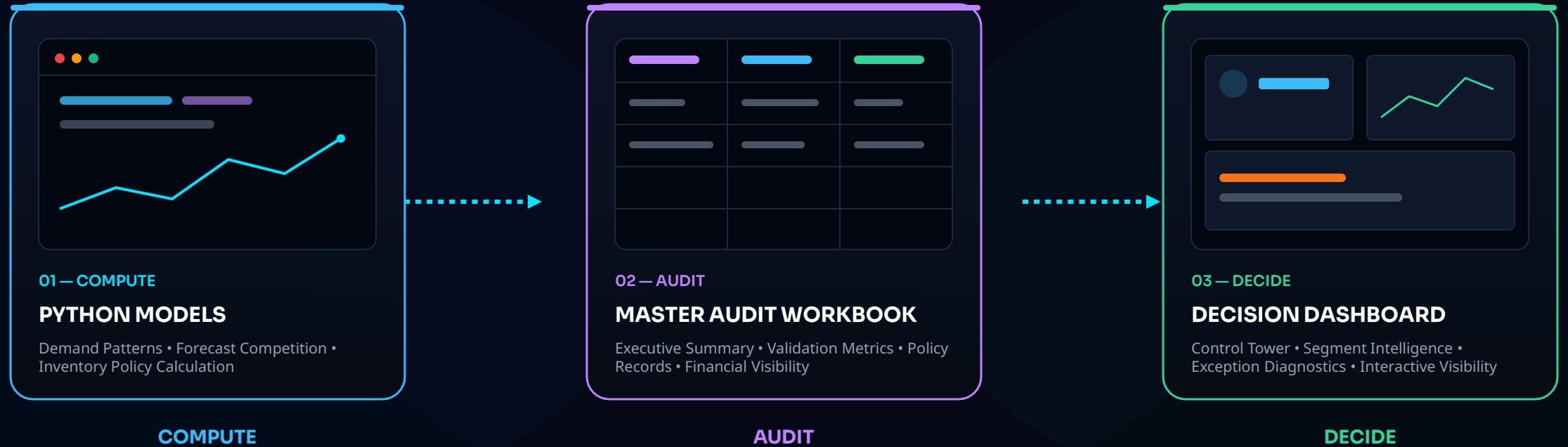
Master Audit Workbook

The auditable record connecting SKU-level analytics to final decisions.



Live Research Demonstration

From analytical computation to auditable decision support



One analytical pipeline. Three decision layers.

The live demonstration validates how the project moves from raw demand data to management-ready decision support.

LIVE DEMONSTRATION

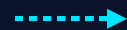
What the Mathematics Means Operationally

How statistical outputs become replenishment decisions

CHAIN 1 — SAFETY STOCK

Higher RMSE

Greater forecast error



Greater Uncertainty

Wider demand uncertainty



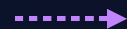
Higher Safety Stock

Larger protection buffer

CHAIN 2 — REORDER POINT

Higher Forecast

Higher expected demand



Higher Lead-Time Demand

More demand in 7-day lead time

7-DAY LEAD TIME



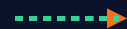
Higher Reorder Point

Earlier replenishment trigger

CHAIN 3 — ORDER QUANTITY (EOQ)

Demand • Ordering Cost • Holding Cost

Key operational & financial inputs



EOQ Calculation

Trade-off optimisation node



Balanced Order Quantity

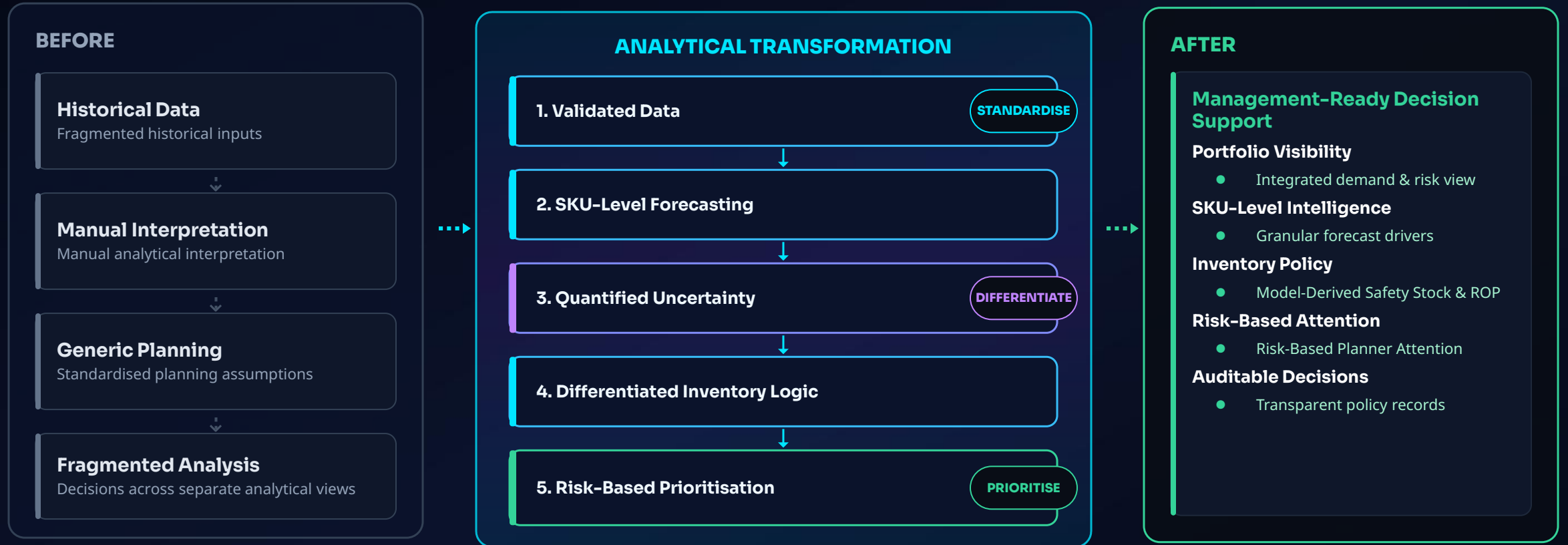
Balances ordering frequency and holding cost

The model converts statistical uncertainty into a measurable replenishment policy.

Forecast performance influences inventory buffers. Demand influences replenishment timing. Cost assumptions influence order quantity.

What the Project Changes for Business Decision-Making

From fragmented historical inputs to a repeatable, auditable decision process

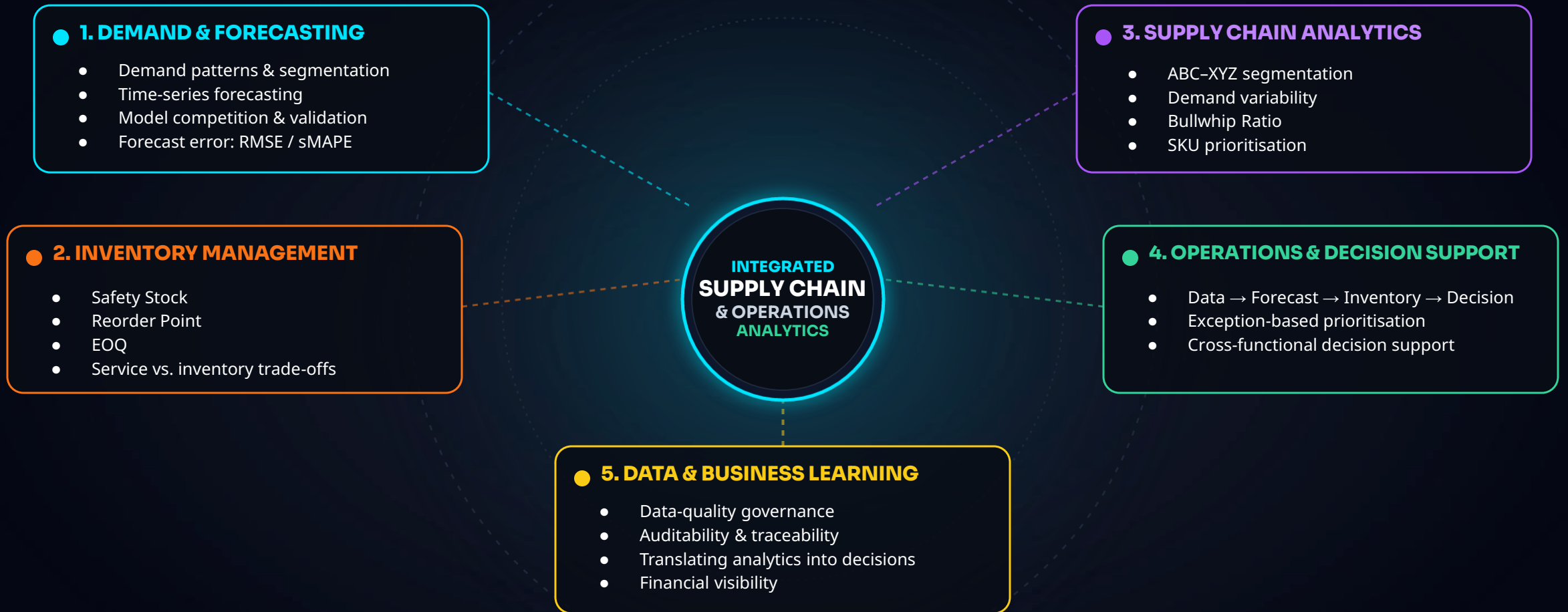


The demonstrated change is analytical: moving from fragmented historical data to a repeatable, auditable decision process.

Realised financial and operational impact requires deployment and post-deployment measurement.

Concepts Learned & Applied — Supply Chain & Operations

Connecting classroom concepts with an enterprise-scale analytical problem



“Classroom concepts become valuable when they can be connected, applied and translated into better decisions.”

Key Learnings & Future Scope — Supply Chain Evolution

"Analytical foundation established. Next: wider supply chain integration."

CURRENT FOUNDATION

WHAT WE LEARNED

- 1 Demand is Heterogeneous**
Different SKUs exhibit different demand behaviour.
- 2 Model Competition Matters**
The most complex model is not automatically the best model.
- 3 Forecasting Connects to Inventory**
Forecast error can be translated into Safety Stock and Reorder Point.
- 4 Data Quality is Foundational**
Reliable inputs are necessary for reliable decisions.

FUTURE STATE — NOT CURRENTLY IMPLEMENTED

WHAT COMES NEXT

- 1 Supplier Intelligence**
Upstream visibility into lead times, supplier reliability & MOQ constraints.
- 2 Live Inventory Integration**
ERP/WMS integration for near-real-time stock position tracking.
- 3 External Demand Signals**
Incorporating market trends, pricing changes & promotional activity.
- 4 Continuous Learning**
Closed-loop feedback for drift detection & continuous model recalibration.

The next opportunity is to connect forecast intelligence with suppliers, live inventory, external demand signals and continuous learning.



THANK YOU